CAP High School Prize Exam

7 April 2006

Marker's Information Sheet

The following information will be used to inform competitors and schools of the exam results, to determine eligibility for some subsequent competitions, and for statistical purposes. Only the marking code, to be assigned by the local examination committee, will be used to identify papers for marking.

Marking Code:

Sponsored by:

Canadian Association of Physicists
Canadian Chemistry and Physics Olympiads

Canadian Association of Physicists 2006 Prize Exam

This is a three-hour exam. National ranking and prizes will be based on a student's performance on sections A, B, and C of the exam. Performance on the questions in parts A and B will be used to determine whose written work in part C will be marked for prize consideration by the CAP Exam National Committee. Part A consists of twenty multiple-choice questions; part B consists of five questions that require graphic solution. The problems in part C can also require graphing.

Non-programmable calculators may be used. The multiple-choice questions must be **on the answer card/sheet** provided. Solutions to the three long problems must be written on **three separate** sheets, as they will be marked by people in different parts of Canada.

Data

Speed of light $c = 3.00\text{x}10^8$ m/s
Gravitational constant $G = 6.67\text{x}10^{11}$ N·m^2/kg^2
Acceleration due to gravity $g = 9.80$ m/s^2
Fundamental charge $e = 1.60\text{x}10^{-19}$ C
Mass of electron $m_e = 9.11\text{x}10^{-31}$ kg
Mass of proton $m_p = 1.67\text{x}\ 10^{-27}$ kg
Planck's constant $h = 6.63\text{x}10^{-34}$ J·s
Coulomb's constant $1/(4\pi\varepsilon_0) = 8.99\text{x}10^9$ N·m^2/C^2
Boltzmann constant $k = 1.38\text{x}10^{-23}$ J/K

Part A: Multiple Choice

Question 1

A meter stick is supported on a fulcrum at the 25-cm mark. A 0.50-kg object is hung from the 0-cm end of the meter stick, and the stick is balanced horizontally. The mass of the meter stick is:

(a) 0.25 kg;
(b) 0.50 kg; *
(c) 0.75 kg;
(d) 1.0 kg

Solution

Condition of equilibrium for torques with respect to the fulcrum is: 0.50x0.25 =Mx0.25→M=0.50 kg

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Question 2

The radius of the Earth is $6.37\cdot10^6$m. Approximately, how fast is a person on the Equator moving due to the Earth rotation?

1. 5m/s
2. 50m/s
3. 500m/s *
4. 5000m/s

Solution

The relationship between the angular speed (ω) and the linear speed (v) gives: $v = \omega R$, where R is the radius of the Earth. If T is a period of rotation, $\omega = 2\pi / T$, and T = 24 x 3600 s.

$v = 2\pi R / T = 4.63\text{x}10^2 \approx 500$ m/s

Question 3

In a certain region of space, the electric field is zero. From this we can conclude that the electric potential in this region is:

(a) constant; *
(b) zero;
(c) positive;
(d) negative.

Solution

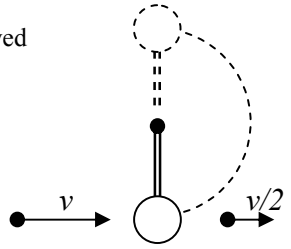
In general, only (a) is the correct answer, because in the absence of field its work on a charge is zero; work is charge multiplied by the potential difference. Therefore,

the potential difference must be zero, and the potential is everywhere the same.

Question 4

A bullet of mass m and speed v hits a pendulum bob of mass M at time t_1, and passes completely through the bob. The bullet emerges at time t_2 with a speed of $v/2$. The pendulum bob is suspended by a stiff rod of length l and negligible mass. After the collision, the bob can barely swing through a complete vertical circle. At time t_3, the bob reaches the highest altitude.

What quantities are conserved in this process?



(a) Total kinetic energy of the bob and the bullet during the time interval $\Delta t = t_2 - t_1$.

(b) Total momentum of the bob and the bullet during the time interval $\Delta t = t_2 - t_1$. *

(c) Total mechanical energy of the bob and the bullet during the time interval $t_3 - t_1$.

(d) Momentum of the bob after t_2.

Solution

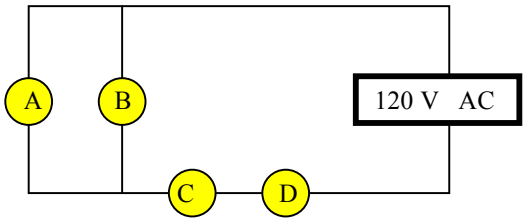
(a) and (c) are wrong, because of the energy loss during the motion of the bullet inside the bob;

(d) is wrong, because an object experiences the external force of gravity that changes momentum.

(b) is the correct answer.

Question 5

The diagram shows a circuit with four identical light bulbs.



When we remove bulb A from the circuit, the light intensity of bulb C:

(a) remains the same;
(b) increases;
(c) decreases;*
(d) becomes zero.

Solution

The equivalent resistance of the part of circuit with the bulb B increases after the removal of the bulb A. This entails the reduction in current through the bulb C.

Question 6

The amplitude of an electromagnetic wave in vacuum is doubled with no other changes made to the wave. As a result of this doubling of the amplitude, which of the following statement is correct?

(a) The speed of propagation of the wave changes.
(b) The frequency of the wave changes.
(c) All of the above are true.
(d) None of the above is true.

(d) is correct because an amplitude, a speed of propagation, and a frequency of the wave are independent variables.

Question 7

Two objects are initially at rest on a frictionless surface. Object 1 has a greater mass than object 2. The same constant force starts to act on each object. The force is removed from each object after it accelerates over the distance d. After the force is removed from both objects, what statement is correct (p is a momentum; K is a kinetic energy)?

(a) $p_1 < p_2$; (b) $p_1 > p_2$; * (c) $K_1 > K_2$; (d) $K_1 < K_2$.

Solution

(b) Object 2 has a greater acceleration because of its smaller mass. Therefore, it takes less time to travel the distance d. Even though the force applied to objects 1 and 2 is the same, the change in momentum is less for object 2 because Δt is smaller.

Question 8

A dart is loaded into a spring-loaded toy dart gun by pushing the spring in by a distance d. For the next loading, the spring is compressed a distance $2d$. How much faster does the second dart leave the gun compared to the first?

(a) four times as fast;
(b) two times as fast;*
(c) the same;
(d) half as fast;
(e) one-fourth as fast.

Solution

According to the law of conservation of energy, the potential energy of a spring transforms into the kinetic energy of a dart. The potential energy is proportional to the square of the compression, and kinetic energy is proportional to the square of the initial speed of the dart. Therefore, the speed increases the same number of times as the compression does.

Question 9

One of two parallel metallic plates is uniformly charged with the charge $+q$, and the other one is charged with the charge $-q$. In this case, the electric field between them is E. When the negatively charged plate is recharged with a positive charge $4q$, the electric field between the plates becomes:

(a) 0;
(b) 1.5 E;*
(c) 2.5 E;
(d) 3E;
(e) 5E.

Solution

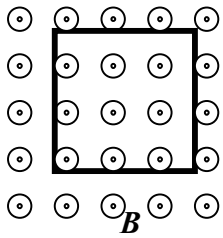
Initially, the field between two plates was $E_i = q / (A\varepsilon_0)$, where A is the area of the plate, and ε_0 is the permittivity of free space. This was a result of addition of the fields of two plates. After the replacement of the negatively charged plate, the field can be calculated as:

$$E_f = E_1 + E_2 = \frac{q}{2A\varepsilon_0} - \frac{4q}{2A\varepsilon_0} = -\frac{3q}{2A\varepsilon_0} = -1.5E_i$$

The magnitude of the field increases by the factor 1.5.

Question 10

A uniform magnetic field ***B*** is directed out of the page. A metallic wire has the shape of a square frame and is placed in the field as shown. The shape of the wire is steadily transformed into a circle in the same plane.



The current in the frame:

(a) is directed clockwise;*
(b) does not appear;
(c) is directed counterclockwise.

Solution

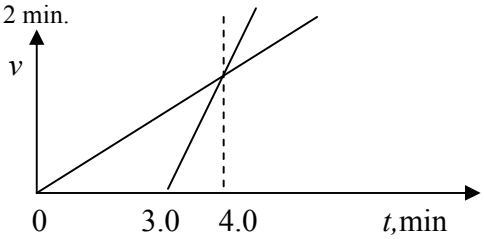
The area of a circle is greater than the area of a square with the same perimeter. During the transformation of the shape of the frame, the magnetic flux through the frame is increasing. It results in the induction of electric current in the loop. According to the Lenz's law, the current produces the magnetic field of the opposite direction to the external field whose flux is increasing. Therefore, the direction of the induced current is clockwise.

Question 11

The drawing shows velocity (v) versus time (t) graphs for two cyclists moving along the same straight segment of a highway from the same point. The second cyclist starts moving at t = 3.0 min.

At what time do the two cyclists meet?

(a) 4.0 min;
(b) 6.0 min;*
(c) 12 min.



Solution

From the slopes on the graph, we can obtain the ratio of accelerations of the two cyclists:

$a_2/a_1 = 4$. The distances traveled by the two cyclists must be equal:

$$\frac{a_1 t^2}{2} = \frac{a_2 (t-3)^2}{2}; \frac{a_2}{a_1} = \left(\frac{t}{t-3}\right)^2;$$

$$\frac{t}{t-3} = 2; \Rightarrow t = 6\,\text{min}$$

Question 12

Two solid spheres manufactured of the same material freely fall down in the air. One sphere has a diameter twice as large as the other does. The force due to air resistance is proportional to the cross-sectional area of a moving object and is a quadratic function of the speed of an object. In some time after the beginning of motion in the presence of air resistance, the velocity of each sphere becomes constant. It is called the terminal velocity. The ratio of terminal velocities of the spheres v_{big}/v_{small} is:

(a) 2; (b) $\sqrt{2}$;* (c) $\frac{1}{2}$; (d) $\frac{1}{\sqrt{2}}$

Solution

The uniform motion with the terminal velocity is possible when the force of gravity of the object is equal to the force of the air resistance:

$$Mg = kv^2 \frac{\pi d^2}{4}; \qquad D\frac{\pi d^3}{6} g = kv^2 \frac{\pi d^2}{4} \quad (1)$$, where

M is the mass of a sphere; d is its diameter; D is the density of the material of the sphere; v is the velocity; and k is the constant coefficient for both spheres. From the equation (1), v is proportional to $\sqrt{d}$.

Question 13

Astronauts are weightless in orbit because:

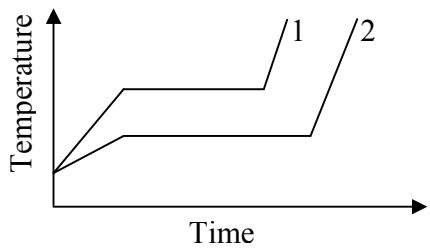
(a) they are beyond the pull of gravity;
(b) they travel in the accelerated frame of reference where fictitious centrifugal force is equal to centripetal force, and the net force on an astronaut is zero;
(c) they are in free fall together with their spacecraft;*
(d) of another cause (specify).

Solution

The single force acting on the spacecraft and the astronauts inside is the force of gravity. Therefore, astronauts are in the free fall.

Question 14

Two solid objects of the same mass are supplied with heat at the same rate $\Delta Q / \Delta t$. The temperature of the first object with latent heat L_1 and specific heat capacity c_1 changes according to the graph 1 on the diagram. The temperature of the second object with latent heat L_2 and specific heat capacity c_2 changes according to the graph 2 on the diagram.



Based on what is shown on the graph, the latent heats L_1 and L_2, and the specific heat capacities c_1 and c_2 in solid state obey the following relationships:

(a) $L_1 > L_2$; $c_1 < c_2$;
(b) $L_1 < L_2$; $c_1 < c_2$;*

(c) There is no information to obtain the relationship between latent heats;
(d) $L_1 > L_2$; $c_1 > c_2$;
(e) $L_1 < L_2$; $c_1 > c_2$.

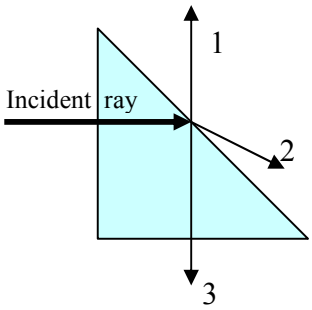
Solution

(b) The horizontal segment of the graph corresponds to the melting of the solids. Under the same condition, the two identical masses have different time of phase transition. The time of the second object is greater, therefore, its latent heat is also greater.

(e) Comparing the slopes of the diagrams for the solid state of the two objects, we see that the same heat causes less change of temperature for the second object. Hence, the second object has the greater specific heat capacitance.

Question 15

A light ray strikes a prism as shown in the drawing. The angles of the prism are 90°, 45°, and 45°. The critical angle of the prism material is 49°. What rays are the possible continuations of the incident ray?



1. The ray 1 only.
2. The ray 2 only.
3. The ray 3 only.
4. Both rays 1 and 3.
5. Both rays 2 and 3.*

Solution

The angle of incidence of the initial ray on the inclined side of the prism is less than the critical angle, therefore, the Snell's law gives the possibility for the ray to be refracted in the direction of the ray 2. In addition, there is also a possibility of reflection along ray 3.

Question 16

You stand on a platform at a train station and listen to a train horn as the train approaches the station at a constant velocity. While the train approaches, but before it arrives, you hear:

(a) the intensity and the frequency of the sound both increasing;
(b) the intensity and the frequency of the sound both decreasing;
(c) the intensity increasing and the frequency decreasing;
(d) the intensity decreasing and the frequency increasing;
(e) the intensity increasing and the frequency remaining the same;*
(f) the intensity and the frequency remaining the same.

Solution

The intensity of sound is increasing because the train approaches. However, the frequency does not change if the speed of the train is the same. Due to the Doppler's effect this frequency is shifted.

Question 17

A linear conductor with current I_1 is placed along the axis of a circular conductor, which carries current I_2.
The magnetic force acting on each of the conductors is:

(a) zero;*
(b) directly proportional to the product of currents I_1 and I_2, and inversely proportional to the radius of the circular conductor;
(c) directly proportional to the product of currents I_1 and I_2, inversely proportional to the square of the radius of the circular conductor;
(d) directly proportional to the product of current I_1, current I_2, and the area of a circular conductor;

Solution

Each conductor is parallel to the direction of magnetic field lines of another conductor. If the wire with current is parallel to the magnetic field line, the sine of the angle between this two directions is zero, and the magnetic force on the wire is zero.

Question 18

An aquarium is filled with water. The lateral wall of the aquarium is 40 cm long and 30 cm high. Using 10 m/s^2 for the acceleration due to gravity, and 1 g/cm^3 for the density of water, the force on the lateral wall of the aquarium is:

(a) 36 N;
(b) 90 N;
(c) 180 N;*
(d) 1500 N.

Solution

The pressure P of water on the base of the aquarium is given by: $P = DgH$, where D is the density of water; H is the height of the wall. Pressure is a linear function of the height. Therefore, the average pressure is half of the maximum one. The force on the lateral wall can be found from: $F = P_{av}HL$, where L is the length of the wall.

$$F = [(1000\text{x}10\text{x}0.3) / 2] \text{ x } 0.3 \text{ x } 0.4 = 180 \text{ N}.$$

Question 19

A satellite is moved from one circular orbit around the earth, to another of lesser radius. Which of the following is true?

(a) The kinetic energy increases and the potential energy increases;
(b) The kinetic energy increases, and the potential energy decreases;*
(c) The kinetic energy decreases and the potential energy decreases;
(d) The kinetic energy decreases and the potential energy increases.

Solution

In the circular motion around the Earth, the centripetal force on the satellite is a gravitational force. Therefore,
$v^2 = GM/R$, where M is the mass of the Earth, R is the radius of the Earth G is the universal gravitational constant. Therefore, the kinetic energy increases with the decrease in the radius of the orbit. The gravitational potential energy is negative and decreases with the decrease in radius.

Question 20

A little particle carries a positive charge $+q$ and is maintained at an altitude h over a large conductive uncharged plate. The electric force of interaction between the charge and the plate is **better** described as:

(a) zero force;
(b) an attractive force proportional to q/h^2;
(c) an attractive force proportional to q^2/h^2;
(d) an attractive force proportional to $q^2/4h^2$;*
(e) a repulsive force proportional to q^2/h^2.

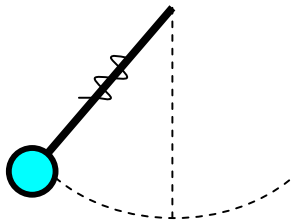
Solution

The lines of field of the point charge are everywhere perpendicular to the surface of the conductive plane. The pattern of the electric field lines over the plane is absolutely the same as for the system of two identical and oppositely charged particles separated by the distance $2h$. Therefore, we can apply the Coulomb's law to calculate the force.

Part B: Questions that require graphical solutions

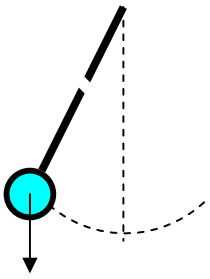
Question 21

A heavy pendulum bob is swinging back and forth when the string, supporting it, suddenly breaks. Ignoring the mass of the string and air resistance, draw the path of the subsequent motion of the bob if the string breaks when the bob is at its highest point.



Solution

The motion will be vertically down under the force of gravity. No other forces are applied to the bob.

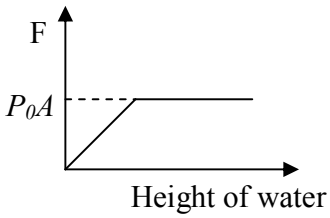


Question 22

A long vertical tube with a weightless movable piston inside has its lower end under the water. The tube is motionless throughout the experiment. The cross-sectional area of the piston is A. The initial position of the piston is just over the surface of the water. Then it is slowly moved upwards. Sketch a graph of the force applied to lift the piston during its motion as a function of the height of water under the piston.

Solution

The force will increase linearly until the pressure of water under the piston becomes equal to the atmospheric pressure P_0. After this moment, the force becomes constant.

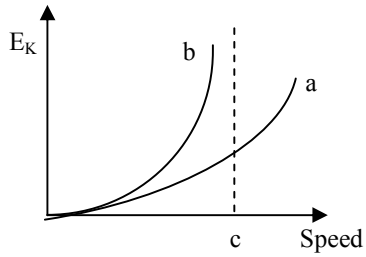


Question 23

On the same graph, draw the following two functions: 1) kinetic energy versus speed of a particle according to classical mechanics and 2) the same function in relativistic theory.

Solution

$c = 3\times10^8$ m/s



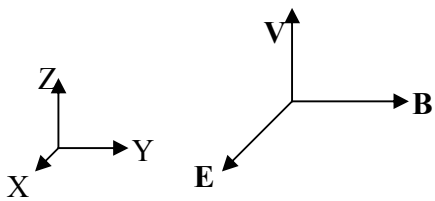
(a) Classical Mechanics (b) Relativistic Mechanics

Question 24

An electron is moving at a constant velocity in a region of space with a uniform electric field of $1.0\cdot10^3$ N/C and a uniform magnetic field of $1.0\cdot10^{-3}$T. The electric field has only an x-component while the magnetic field has only a y-component.

Sketch a vector diagram for the velocity of the electron, magnetic field, and electric field in the Cartesian system of coordinates.

Solution

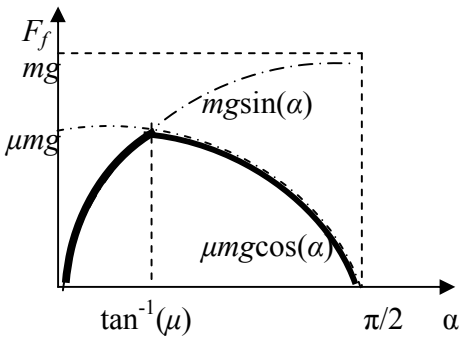


Question 25

A brick is on an incline whose angle of inclination can be changed from 0 to 90 degrees.

Sketch a graph of the force of friction acting on the brick versus the angle of inclination in the given range.

Solution

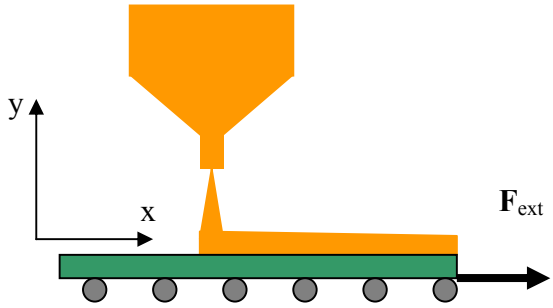


There are two segments of the graph: 1) when the brick is motionless, and static friction exists; and

2) when the brick moves and the force of kinetic friction acts on it. The motion starts when $\alpha=\tan^{-1}(\mu)$

1) For the motionless brick: $F_f = mg\sin(\alpha)$
2) For the moving brick: $F_{fk} = \mu mg\cos(\alpha)$

Part C: Problems
Problem 1



Sand from a stationary hopper falls onto a moving conveyor belt at a rate of 5.00 kg/s as in Fig.1. The conveyor belt is supported by frictionless rollers and moves at a constant speed of 0.750 m/s under the action of a constant horizontal external force $\mathbf{F}_{ext}$ supplied by the motor that drives the belt. Find:

1) the force of friction exerted by the belt on the sand;
2) the external force $\mathbf{F}_{ext}$;
3) the work done by $\mathbf{F}_{ext}$ in 1 s;
4) the kinetic energy acquired by the falling sand each second due to the change in its horizontal motion;
5) compare and analyse the answers to 3) and 4).

Solution

Introducing the vertical and horizontal axes of coordinates, we can find the rate of change of the x-component of momentum for the sand:

(1) $\dfrac{\Delta p_x}{\Delta t} = \dfrac{(5.00\ \text{kg})(0.750\ \text{m/s})}{1.00\ \text{s}} = \boxed{3.75\ \text{N}}$

This change occurs due to the only horizontal force on the sand – the force of friction: $f = \dfrac{\Delta p_x}{\Delta t} = \boxed{3.75\ \text{N}}$

(2) The belt is in equilibrium. It experiences the external force and the force of friction from the sand:

$\sum F_x = m a_x: \quad +F_{ext} - f = 0$ and $F_{ext} = \boxed{3.75\ \text{N}}$

(3) $W = F\Delta r\cos\theta = 3.75\ \text{N}\,(0.750\ \text{m})\cos 0° = \boxed{2.81\ \text{J}}$

(4) $\frac{1}{2}(\Delta m)v^2 = \frac{1}{2}5.00\ \text{kg}(0.750\ \text{m/s})^2 = \boxed{1.41\ \text{J}}$

(5) Friction between the sand and the belt converts half of the input work into internal energy of both objects.

Problem 2

A source of electric power with electromotive force E and internal resistance r, is connected to a resistor.

1) Find the power delivered to the resistor as a function of the resistance of the resistor.
2) Find the power produced by the source as a function of the resistance of the resistor.
3) Sketch graphs of these functions.
4) Treating the efficiency of the source as a ratio of the consumed power to the power produced by the source, find the efficiency of the source of electric power as a function of the resistance of the resistor and sketch the graph of this function.
5) Find the resistance of the resistor that corresponds to the maximum power delivered to the resistor, and calculate the efficiency for this resistance.

Solution

1)The power, delivered to a resistor, is given by:

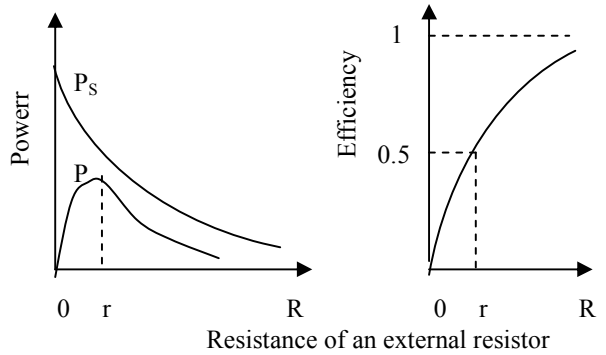
$P = \dfrac{E^2}{(R+r)^2}R$. This function has a maximum at $R = r$.

$P_{max} = E^2/(4r)$.

2) The power, produced by the source is :

$P_S = \dfrac{E^2}{(R+r)^2}(R+r) = \dfrac{E^2}{(R+r)}$. This function is steadily decreasing with increasing of the resistance R (when $R = 0$, $P_S = E^2/r$).

4) The efficiency is given by: $e = R/(R+r)$



Problem 3

A point source of light is moving uniformly along a straight line, which intersects the optical axis of a converging lens under a small angle α at a distance 2F from the lens with a focal distance F. The speed of the source of light is v_0.

i. Draw the lens, the trajectory of the source of light, and the trajectory of its image produced by the lens.
ii. Draw the vector diagram for the velocity of the source of light, the velocity of the image, and the relative velocity of the image with respect to the source.
iii. Determine the minimum relative speed of the image of the source of light with respect to the source.

Solution

First, we must investigate the trajectory of the image of the source of light. The path AB of the source during some time interval is shown on the diagram (fig 3.1) as an object. Its image A'B' is a trajectory of the image of the point source of light during the same time interval.

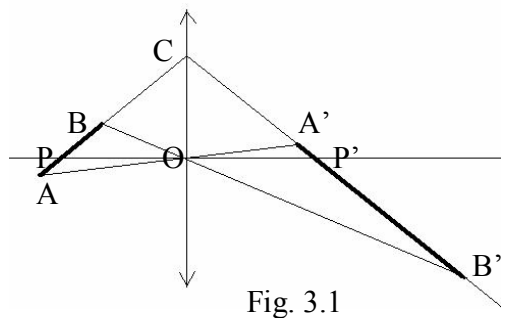


Fig. 3.1

Due to the properties of the point at the double-focal distance from the lens, the image of the object is also at the double-focal distance from the lens. Therefore, the trajectory of the image of the source of light A'B' crosses the optical axis at the point P' at the distance 2F from the lens. The triangles POC and P'OC are equivalent right angle triangles, and thus, the trajectory of the source and the trajectory of its image cross the optical axis of the lens under the same angle α. The diagram shows the different distances AB and A'B', covered by the source and its image during the same time interval, which permits to state that the speeds of the source and the image are different.
While the object moves at the velocity $\boldsymbol{v_0}$, its image moves at the velocity $\boldsymbol{u}$ along the line that crosses the optical axis at distance 2F from the vertex of lens and has the same magnitude of inclination α to the axis (fig.3.2).

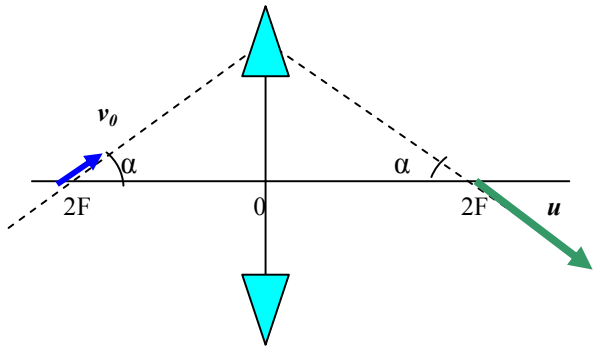


Fig.3.2

The vector diagram of the velocities (fig.3.3) shows that the relative velocity $\boldsymbol{V}$ has its minimum magnitude when it is perpendicular to the vector $\boldsymbol{u}$. From this condition, we can obtain the result: $|\boldsymbol{V}|_{\min} = v_0 \sin 2\alpha$.

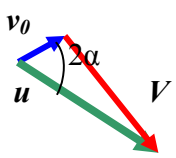


Fig.3.3

The obtained condition, however, does not permit us to conclude about the instantaneous position of the source of light at the moment that matches the result for the relative velocities, as it is clear from fig. 3.1. The only general conclusion is that the position of the source must guaranty the following relationship between the instantaneous speed of the source and the instantaneous speed of its image:

$$v_0 < u$$